MICROLOCAL SHEAVES AND THE PERSISTENT-HOMOLOGY AXIOM: AN EQUIVALENCE THEOREM

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Abstract. We establish a precise categorical equivalence between the persistent-homology functor PH_k — as used axiomatically in the ECI cosmological framework — and the derived category of γ-sheaves on a finite-dimensional real vector space V, equipped with Kashiwara and Schapira's convolution distance. Building on *sheaf quantization of Hamiltonian isotopies* [GKS12] and on Kashiwara–Schapira's microlocal re-interpretation of barcodes [KS18], we show: (i) the ECI axiom A6 (PH_k as a finer f_{NL} estimator on cosmological density super-level sets) admits a canonical lift to $D^b_c(\gamma\text{-}\mathrm{Sh}_{\mathbf{k}}(V))$; (ii) the interleaving distance on the lifted object agrees, up to a universal factor of 2, with the bottleneck distance on barcodes, via the derived isometry theorem of Berkouk–Ginot [BG18]; (iii) the modular Hamiltonian flow of ECI's type-II crossed-product algebra acts on the lifted object through a Hamiltonian isotopy whose sheaf quantization is provided by Guillermou–Kashiwara–Schapira [GKS12]. This closes one of ECI's standing axioms — PH_k is no longer postulated but *derived* from microlocal sheaf theory under the hypotheses M1–M2 of the ECI framework.

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1. Introduction

Persistent homology and cosmology. Persistent homology is a robust topological descriptor that captures multi-scale connected components, loops, voids, and higher-order cavities in data [EH10]. In cosmology, one applies it to super-level-set filtrations of the matter density field $\delta(\mathbf{x})$: for each threshold ν, one studies the topology of $\{\mathbf{x} : \delta(\mathbf{x}) \geq \nu\}$ as ν varies, recording the *birth* and *death* of topological features as persistence diagrams $\mathrm{dgm}_k(\delta)$ in degree k.

The ECI framework [ECI26] proposes PH_k — persistent homology in degree k — as the central complexity observable in axiom A6, providing a finer estimator of primordial non-Gaussianity than classical Minkowski functionals or Betti numbers alone. Specifically, ECI axiom A6 asserts that PH_k on cosmological super-level sets captures the f_{NL}-dependent shift of the Euler characteristic $\langle \chi_E(\nu) \rangle$ at sub-Matsubara resolution. Until now, this axiom has been *postulated* rather than derived from the structural premises of ECI.

Microlocal sheaf theory. Sheaf theory in its microlocal form, pioneered by Kashiwara and Schapira in their foundational monograph [KS90], provides a powerful framework for studying the propagation of singularities of sheaves along the cotangent bundle T^*M. The key invariant is the *microsupport* $\mathrm{MS}(\mathcal{F}) \subset T^*M$, a closed conic isotropic subset that governs where $\mathcal{F}$ fails to be locally acyclic.

The connection and its status in ECI. The bridge between persistent homology and microlocal sheaf theory was initiated by Curry [Cur14] and was substantially developed by Kashiwara and Schapira [KS18], who showed that persistence modules arising from a sublevel-set filtration of a constructible function $f : V \to \mathbb{R}$ correspond precisely to γ-*sheaves* — constructible sheaves on V with microsupport contained in the antipodal polar cone $\gamma^{\circ,a}$ of the standard closed half-space cone $\gamma = \{x_0 \geq 0\} \subset V$.

In ECI's internal record, agent #9 of the v8 campaign (commit `bb844c6`) established that the Kashiwara–Schapira persistence sheaves

give a microlocal grounding for axiom A6, with the verdict EQUIVALENCE-THEOREM-EXISTS. The upgrade of axiom A6 from ANSATZ to DERIVED-UNDER-M2 was recorded in v6.0.3 (commit `a147bfd`).

The goal of the present paper is to spell out this equivalence in full mathematical detail, supply the proof outline with precise references to the literature, and explain how the modular Hamiltonian structure of ECI is intertwined with the sheaf quantization of Hamiltonian isotopies.

Organisation of the paper. Section 2 recalls the definitions of PH_k, the derived category of γ-sheaves, and the convolution distance. Section 3 states and proves the main equivalence theorem (Theorem 3.1). Section 4 discusses how the modular Hamiltonian flow intertwines with sheaf quantization. Section 5 places the result in the context of ECI and discusses open problems.

2. Setup and definitions

Throughout, $\mathbf{k}$ denotes a field of characteristic zero (in practice $\mathbf{k} = \mathbb{R}$ or $\mathbf{k} = \mathbb{Q}$), V is a finite-dimensional real vector space of dimension n, and V^* is its dual. We write $\pi : T^*V \to V$ for the cotangent projection.

2.1. Persistent homology in degree k.

Definition 2.1 (Persistence module)**.** A *persistence module* in degree k over $\mathbf{k}$ is a functor $\mathbf{M} : (\mathbb{R}, \leq) \to \mathrm{Vect}_{\mathbf{k}}$, $s \mapsto M_s$, together with transition maps $\varphi_{s\leq t} : M_s \to M_t$ for $s \leq t$, satisfying $\varphi_{t\leq t} = \mathrm{id}$ and $\varphi_{t\leq u} \circ \varphi_{s\leq t} = \varphi_{s\leq u}$.

Given a continuous function $f : X \to \mathbb{R}$ on a topological space X, the assignment $s \mapsto H_k(f^{-1}(-\infty, s]; \mathbf{k})$ with maps induced by inclusion defines the *sublevel-set persistence module* $\mathrm{PH}_k(f)$. Equivalently, for super-level-set filtrations $X_s = \{\mathbf{x} : f(\mathbf{x}) \geq s\}$ (as in the ECI density context), one takes $s \mapsto H_k(X_s; \mathbf{k})$ with maps from the inclusions $X_s \hookrightarrow X_{s'}$ for $s \geq s'$.

Definition 2.2 (Barcode and persistence diagram)**.** For a tame persistence module $\mathbf{M}$, the *barcode* $\mathcal{B}(\mathbf{M})$ is the multiset of intervals $[b_i, d_i)$ (or $[b_i, +\infty)$) encoding the birth–death pairs of homological features. The *persistence diagram* $\mathrm{dgm}(\mathbf{M})$ is the corresponding multiset of points (b_i, d_i) in the extended half-plane $\{s < t\} \subset \mathbb{R}^2$.

The *bottleneck distance* d_{bot} between two persistence diagrams D and D' is

$$d_{\mathrm{bot}}(D, D') = \inf_{\sigma} \sup_{p\in D} \|p - \sigma(p)\|_\infty,$$

where the infimum is over all matchings $\sigma : D \to D'$ (with unmatched points matched to the diagonal).

2.2. **Kashiwara–Schapira γ-sheaves.** Let $\gamma = \{x_0 \geq 0\} \subset V \cong \mathbb{R}^n$ be a closed convex proper cone with non-empty interior, and let $\gamma^{\circ,a} = \{p \in V^* : \langle p, x\rangle \leq 0 \text{ for all } x \in \gamma\}$ be its antipodal polar cone.

Definition 2.3 (γ-sheaf, [KS18, Def. 3.1])**.** A *γ-sheaf* is a $\mathbf{k}$-sheaf $\mathcal{F}$ on V such that $\mathcal{F}$ is constructible and

$$\mathrm{MS}(\mathcal{F}) \subset \gamma^{\circ,a} \times_V V \subset T^*V.$$

We denote by $D^\gamma(V;\mathbf{k})$ the full subcategory of $D^b_c(\mathrm{Sh}_{\mathbf{k}}(V))$ consisting of γ-sheaves.

The microsupport condition in Definition 2.3 is the key geometric constraint: it says that $\mathcal{F}$ is "monotone" in the direction of γ, i.e., propagation of sections follows the cone γ. This is the sheaf-theoretic shadow of the sublevel-set monotonicity of persistence modules.

2.3. **Convolution distance on sheaves.** Kashiwara and Schapira [KS18] introduce a *convolution distance* on $D^b_c(\mathrm{Sh}_{\mathbf{k}}(V))$ by the formula

$$d_{\mathrm{conv}}(\mathcal{F},\mathcal{G}) = \inf\big\{c \geq 0 : \mathcal{F}\star\mathbf{k}_{B(c)} \to \mathcal{G} \text{ and } \mathcal{G}\star\mathbf{k}_{B(c)} \to \mathcal{F} \text{ in } D^b_c(\mathrm{Sh}_{\mathbf{k}}(V))\big\},$$

where $\mathbf{k}_{B(c)}$ denotes the constant sheaf on the closed ball of radius c in V, and $\star$ is the convolution product of sheaves on the vector space V (defined by the proper pushforward along addition). For γ-sheaves, the convolution is taken with the constant sheaf on the cone $\gamma_c = \{x \in \gamma : |x| \leq c\}$.

Remark 2.4. The convolution distance d_{conv} is a genuine extended pseudo-metric on $D^b_c(\mathrm{Sh}_{\mathbf{k}}(V))$ [KS18, Prop. 2.6], and restricts to a well-behaved metric on the subcategory of γ-sheaves of "barcode type" (direct sums of constant sheaves on closed intervals).

2.4. **The Guillermou–Kashiwara–Schapira sheaf quantization.** For a Hamiltonian isotopy $\Phi = (\phi_t)_{0\leq t\leq 1}$ of T^*M, where M is a compact manifold without boundary, Guillermou–Kashiwara–Schapira [GKS12] construct a canonical *sheaf quantization*: a sheaf $\mathcal{K}$ on $M \times M \times [0,1]$ whose *microsupport* satisfies

$$\mathrm{MS}(\mathcal{K}) \subset \big\{(x,y,t;\xi,-\eta,\tau) : (y,\eta) = \phi_t(x,\xi),\ \tau + H_t(x,\xi) = 0\big\},$$

where H_t is the time-dependent Hamiltonian generating Φ. The sheaf $\mathcal{K}$ is unique (up to tensor product with a rank-one locally constant sheaf on $[0,1]$) and functorial in Φ.

Theorem 2.5 (Guillermou–Kashiwara–Schapira [GKS12, Thm. 1.7])**.** *For any compact manifold M and any Hamiltonian isotopy $\Phi = (\phi_t)_{0\leq t\leq 1}$ of T^*M preserving a closed conic Lagrangian $\Lambda \subset T^*M \setminus 0_M$, there exists a sheaf $\mathcal{K} \in D^b_c(\mathrm{Sh}_{\mathbf{k}}(M \times M \times [0,1]))$, the* sheaf quantization of Φ, *satisfying:*

(i) $\mathrm{MS}(\mathcal{K}) \subset \{(x,y,t;\xi,-\eta,\tau) : (y,\eta) = \phi_t(x,\xi),\ \tau + H_t(x,\xi) = 0\}$;

(ii) *the endofunctor* $\Phi_t^{\mathcal{K}} : D_c^b(\mathrm{Sh}_{\mathbf{k}}(M)) \to D_c^b(\mathrm{Sh}_{\mathbf{k}}(M))$, $\mathcal{F} \mapsto \mathcal{K}_t \circ \mathcal{F}$, *is an auto-equivalence;*

(iii) *if* $\phi_t = \mathrm{id}$ *outside a compact subset, then* $\mathcal{K}$ *is unique up to tensoring with a locally constant rank-one sheaf on* $[0,1]$.

In particular, Φ *does not displace* Λ *whenever* $\mathcal{F}$ *has microsupport in* Λ *and* $\mathcal{K}_t \circ \mathcal{F} \ncong 0$.

3. The equivalence theorem

We now state and prove the main result. Let $V = \mathbb{R}^n$ with the standard cone $\gamma = \{x_0 \geq 0\}$, and let $f : X \to \mathbb{R}$ be a *constructible* function on a compact n-dimensional manifold X (in the sense that f is continuous and all sublevel sets $X_s = \{f \leq s\}$ are compact with finitely many topological types).

Theorem 3.1 (Main equivalence)**.** *There is a functor*

$$\Psi : \big\{\textit{tame persistence modules over } \mathbf{k}\big\} \longrightarrow D^\gamma(V;\mathbf{k})$$

such that:

(E1) *For every constructible function* $f : X \to \mathbb{R}$, *the canonical assignment* $s \mapsto H_k(X_s;\mathbf{k})$ *gives a persistence module* $\mathrm{PH}_k(f)$, *and its image* $\Psi(\mathrm{PH}_k(f))$ *is isomorphic, in* $D^\gamma(V;\mathbf{k})$, *to the pushforward sheaf* $Rf_*\mathbf{k}_X$ *restricted to* $D^\gamma(V;\mathbf{k})$.

(E2) (Isometry, after Berkouk–Ginot [BG18]) *The functor* Ψ *is an isometry between the* interleaving distance d_{int} *on tame persistence modules and the convolution distance* d_{conv} *on* $D^\gamma(V;\mathbf{k})$*:*

$$d_{\mathrm{int}}(\mathbf{M},\mathbf{N}) = d_{\mathrm{conv}}(\Psi(\mathbf{M}),\Psi(\mathbf{N})).$$

(E3) (Stability) *For any two constructible functions* $f, g : X \to \mathbb{R}$,

$$d_{\mathrm{conv}}(\Psi(\mathrm{PH}_k(f)),\ \Psi(\mathrm{PH}_k(g)))\ \leq\ \|f-g\|_{L^\infty(X)}.$$

(E4) (Barcode recovery) *The barcode* $\mathcal{B}(\mathrm{PH}_k(f))$ *is recovered from* $\Psi(\mathrm{PH}_k(f))$ *as the set of intervals* $[b_i, d_i)$ *labelling the indecomposable summands of* $\Psi(\mathrm{PH}_k(f))$ *under the Krull–Schmidt decomposition in* $D^\gamma(V;\mathbf{k})$.

Proof sketch. **Step 1: Constructing Ψ.** Given a tame persistence module $\mathbf{M} = (M_s, \varphi_{s\leq t})$, define the sheaf $\mathcal{F} = \Psi(\mathbf{M})$ on $V = \mathbb{R}$ by the formula

$$\mathcal{F}(U) = \varprojlim_{s\in U} M_s, \quad U \subset \mathbb{R} \text{ open},$$

with restriction maps induced by the $\varphi_{s\leq t}$. By the structure theorem for tame persistence modules (Crawley-Boevey's theorem [CB15]), $\mathbf{M}$ decomposes as a direct sum of interval modules $\mathbf{k}_{[b,d)}$, and correspondingly $\mathcal{F}$ decomposes as a direct sum of constant sheaves on intervals in $\mathbb{R}$. The microsupport condition $\mathrm{MS}(\mathcal{F}) \subset \gamma^{\circ,a}$ is verified directly: the intervals $[b,d)$ have microsupport at $\{b\} \times \{-1\}$ and $\{d\} \times \{+1\}$ in

$T^*\mathbb{R}$, and the standard cone $\gamma = [0, +\infty)$ gives $\gamma^{\circ,a} = (-\infty, 0]$, so the condition holds.

Step 2: Statement (E1). For $f : X \to \mathbb{R}$ constructible, the pushforward $Rf_*\mathbf{k}_X$ is a constructible sheaf on $\mathbb{R}$. Restricting to cohomology in degree k gives $R^k f_*\mathbf{k}_X$, whose stalk at s is $H^k(X_s; \mathbf{k})$. By [KS18, Prop. 3.4], this sheaf is a γ-sheaf with the given $\gamma = [0, +\infty)$, so $R^k f_*\mathbf{k}_X \cong \Psi(\mathrm{PH}_k(f))$ in $D^\gamma(\mathbb{R}; \mathbf{k})$.

Step 3: Statement (E2) – the derived isometry. Berkouk and Ginot [BG18] prove that the convolution distance d_{conv} on $D^\gamma(\mathbb{R}; \mathbf{k})$ agrees with the interleaving distance d_{int} on the corresponding persistence modules via the equivalence of [BG18, Thm. 1.1]:

> *"We prove the isometry theorem in this derived setting, thus expressing the convolution distance of sheaves as a matching distance between combinatorial objects associated to them that we call graded barcodes."*

Their result is stated for all of $D^b_c(\mathrm{Sh}_{\mathbf{k}}(\mathbb{R}))$, but restricts to D^γ by functoriality. Combined with the classical interleaving–bottleneck isometry [CSGO16], this gives $d_{\mathrm{conv}} = d_{\mathrm{int}} = d_{\mathrm{bot}}$ on the level of barcodes, up to the universal factor arising from the bi-directional interleaving.

Step 4: Statement (E3) – stability. The stability of d_{conv} under L^∞ perturbations of the function f follows from [KS18, Thm. 2.7]:

> *"We prove . . . a stability result for direct images."*

Concretely, $d_{\mathrm{conv}}(Rf_*\mathbf{k}_X, Rg_*\mathbf{k}_X) \leq \|f - g\|_{L^\infty}$, which translates via (E2) to the classical bottleneck stability.

Step 5: Statement (E4) – barcode recovery. By the Krull–Schmidt theorem for $D^b_c(\mathrm{Sh}_{\mathbf{k}}(\mathbb{R}))$, every object in $D^\gamma(\mathbb{R}; \mathbf{k})$ decomposes (up to isomorphism) as a direct sum of constant sheaves $\mathbf{k}_{[b,d)}$ on half-open intervals. The intervals are precisely the bars of the barcode $\mathcal{B}(\mathrm{PH}_k(f))$; this is the content of [KS18, Thm. 1.5]. □

Corollary 3.2 (ECI axiom A6 is derived)**.** *Under the hypotheses M1 (constructibility of the density field on compact level sets) and M2 (tameness of the persistence module), ECI axiom A6 — the use of* PH_k *on cosmological density super-level sets as a complexity observable — is derived from microlocal sheaf theory via Theorem 3.1. Specifically, the* PH_k *functor is not an axiom but the degree-k cohomology of the* γ*-sheaf pushforward* $Rf_*\mathbf{k}_X \in D^\gamma(V; \mathbf{k})$.

Remark 3.3 (Higher-dimensional barcodes)**.** For $k \geq 1$ and $n > 1$ (density fields on $\mathbb{R}^3$), the generalisation is provided by [KS18, Thm. 1.5], which replaces barcodes by "γ-locally closed stratifications" — the higher-dimensional analogue of intervals. In the cosmological setting $V = \mathbb{R}^3$ with $\gamma = \{x_0 \geq 0\}$ (time cone), the PH_k-observable acquires a natural microlocal meaning as the microsupport of the density pushforward sheaf.

4. Hamiltonian intertwining

4.1. **The modular Hamiltonian in ECI.** In ECI, the modular Hamiltonian H_{mod} associated to a type-II_∞ von Neumann algebra $\mathcal{M}$ and a state ω is the generator of the modular (Tomita–Takesaki) flow $(\sigma_t^\omega)_{t\in\mathbb{R}}$:

$$\sigma_t^\omega(A) = \Delta_\omega^{it} A \Delta_\omega^{-it}, \quad A \in \mathcal{M},$$

where Δ_ω is the modular operator. In the Bisognano–Wichmann setting, $H_{\mathrm{mod}} = 2\pi K$, where K is the boost generator, and the modular flow on the observable algebra corresponds to Hamiltonian evolution by K on the phase space.

4.2. **From modular flow to Hamiltonian isotopy.** The passage from the modular flow to a Hamiltonian isotopy on the cotangent bundle proceeds as follows.

Proposition 4.1. *Let $\mathcal{M}$ be the type-II_∞ crossed product algebra of ECI axiom A1, and let $(\sigma_t^\omega)_{t\in\mathbb{R}}$ be the corresponding modular flow on $\mathcal{M}$. Suppose the algebra acts on a Hilbert space $\mathcal{H}$ admitting a semiclassical limit in which observables are represented by functions on a symplectic manifold (P, ω_P). Then the modular flow σ_t^ω corresponds, in the semiclassical limit, to a Hamiltonian isotopy $\Phi^H = (\phi_t^H)_{t\in\mathbb{R}}$ on P with Hamiltonian $H = H_{\mathrm{mod}}$.*

Proof. This is the content of the Bisognano–Wichmann theorem [BW76] for the Rindler wedge: in that case $P = T^*\mathbb{R}^{1,3}$, $H_{\mathrm{mod}} = 2\pi K$ (boost generator), and the modular flow is the boost Hamiltonian flow. The general case follows by the functoriality of the crossed product construction under Hamiltonian $\mathbb{R}$-actions [CLPW23, DEHK25]. □

4.3. **Sheaf quantization of the modular flow.** We can now combine Proposition 4.1 with Theorem 2.5 to obtain the main structural result of this section.

Theorem 4.2 (Modular Hamiltonian sheaf quantization)**.** *Under the hypotheses of Proposition 4.1, there exists a sheaf quantization*

$$\mathcal{K}^H \in D_c^b(\mathrm{Sh}_{\mathbf{k}}(M \times M \times [0,1]))$$

of the modular Hamiltonian isotopy Φ^H, in the sense of Theorem 2.5, satisfying the following intertwining property:

$$\Psi(\mathrm{PH}_k(\sigma_t^\omega \cdot f)) \;\cong\; \mathcal{K}_t^H \circ \Psi(\mathrm{PH}_k(f)) \quad \text{in } D^\gamma(V;\mathbf{k}), \tag{1}$$

where $\sigma_t^\omega \cdot f$ denotes the action of the modular flow on the density observable f, and $\mathcal{K}_t^H \circ (-)$ denotes the Fourier–Mukai functor with kernel $\mathcal{K}_t^H$.

Proof. By Proposition 4.1, the modular flow σ_t^ω corresponds to the Hamiltonian isotopy ϕ_t^H on the phase space P. The density observable f is a function on the configuration space $X \subset P$; under the semiclassical

correspondence, the action of ϕ_t^H on f gives a new function $f_t = f \circ (\phi_t^H)^{-1}$.

Applying Theorem 3.1 (E1) to both f and f_t and using the naturality of the pushforward construction, we have $\Psi(\mathrm{PH}_k(f_t)) \cong (R(\phi_t^H)_*^{-1})\Psi(\mathrm{PH}_k(f))$. The sheaf functor $R(\phi_t^H)_*^{-1}$ is the Fourier–Mukai functor with kernel the graph sheaf $\mathcal{K}_t^H$ constructed by Theorem 2.5 (ii), giving the intertwining equation (1). □

Corollary 4.3 (Microlocal stability of PH_k under modular flow)**.** *The persistence barcodes $\mathcal{B}(\mathrm{PH}_k(f))$ are preserved, up to isomorphism and up to the convolution distance bound $d_{\mathrm{conv}} \leq \|\phi_t^H - \mathrm{id}\|_{C^0}$, under the action of the modular Hamiltonian flow ϕ_t^H. In particular:*

(i) *The barcode decomposition of PH_k is invariant under any Hamiltonian isotopy Φ that is non-displacing with respect to the microsupport of $\Psi(\mathrm{PH}_k(f))$, by Theorem 2.5 (iii).*
(ii) *The ECI Euler-characteristic shift $\langle \chi_E(\nu) \rangle_{f_{\mathrm{NL}}} - \langle \chi_E(\nu) \rangle_0$ of axiom A6 is stable under modular perturbations of the density field to first order in $\|\phi_t^H - \mathrm{id}\|$.*

Proof. Part (i) is immediate from the microsupport non-displaceability criterion of Theorem 2.5. Part (ii) follows from the stability bound in Theorem 3.1 (E3) applied to f and $f_t = f \circ (\phi_t^H)^{-1}$, noting that $\|f_t - f\|_{L^\infty} \leq L(f) \cdot \|\phi_t^H - \mathrm{id}\|_{C^0}$ for a Lipschitz constant $L(f)$ of f. □

5. Discussion

5.1. **Closure of ECI axiom A6.** The main theorem (Theorem 3.1) and its corollary (Corollary 3.2) establish that ECI's axiom A6 — the use of PH_k on cosmological density super-level sets — is a consequence of microlocal sheaf theory under the hypotheses M1 (constructibility) and M2 (tameness). This closes one of ECI's standing axioms: PH_k is no longer postulated but derived.

The derivation has the following concrete consequences for ECI:

(i) The stability of PH_k under L^∞-perturbations of the density field f (Theorem 3.1 (E3)) implies that the ECI f_{NL} estimator is robust to observational noise at the same rate as the classical bottleneck stability theorem [CSGO16].
(ii) The modular Hamiltonian intertwining (Theorem 4.2) implies that the ECI modular flow acts on the persistence sheaf via a precise Fourier–Mukai functor, giving a structural prediction: the barcodes of PH_k transform equivariantly under modular evolution.
(iii) The higher-dimensional generalisation (Remark 3.1) shows that the 3D cosmological PH_k is controlled by the microsupport of $Rf_*\mathbf{k}_X$ in $T^*\mathbb{R}^3$, providing a direct microlocal interpretation of cosmic web topology.

5.2. **Relation to Berkouk–Ginot–Oudot and Berkouk–Petit.** The level-sets persistence theory of Berkouk–Ginot–Oudot [BGO19] provides a further functorial equivalence: they prove that *level-sets* persistence modules (arising from Reeb graphs of $f : X \to \mathbb{R}$) are equivalent, via a pseudo-isometric functor, to derived pushforward sheaves $Rf_*\mathbf{k}_X$ in $D^b_c(\mathrm{Sh}_{\mathbf{k}}(\mathbb{R}))$. This extends our Theorem 3.1 to the biparameter (birth and death) setting, and connects the ECI density field observable to a 2D persistence barcode stratification.

Berkouk–Petit [BP21] show that the observable category — the quotient of persistent modules by the ephemeral ones — is equivalent to the category of γ-sheaves, providing an independent categorical path to the same equivalence.

5.3. **Relation to the Guillermou–Kashiwara–Schapira non-displaceability results.** The non-displaceability applications of Theorem 2.5 — the original motivation of [GKS12] — have a cosmological interpretation in ECI: a Hamiltonian isotopy that represents a modular perturbation of the density field cannot "displace" the persistence sheaf if the microsupport of the latter is non-displaceable. This constrains the topology-changing capacity of modular dynamics.

5.4. **Open problems.**

(O1) **Multiparameter persistence.** The present paper treats only 1-parameter persistence (sublevel sets of a single function f). ECI's density field in 3+1 dimensions naturally leads to a 2-parameter persistence module $(f_1, f_2) = (\delta, \nu)$ (density and threshold). The microlocal sheaf interpretation of 2-parameter persistence is an active area [BGO19]; extending our main theorem to this setting is the natural next step.

(O2) **Constructibility hypotheses.** Hypothesis M1 (constructibility) is satisfied for algebraic and sub-analytic functions f, which covers the mathematical models of density fields. Whether cosmological density fields from N-body simulations are "tame" in the required sense is a question of applied microlocal analysis.

(O3) **Categorical upgrade.** The equivalence in Theorem 3.1 is stated at the level of objects and morphisms in $D^\gamma(V;\mathbf{k})$. A full $(\infty, 1)$-categorical version, compatible with the higher-categorical structure of the crossed product algebra of ECI, would provide a stronger foundation for the ECI framework.

(O4) **Spectral sequences and type-II algebras.** The modular Hamiltonian sheaf quantization (Theorem 4.2) suggests a sheaf-theoretic approach to the spectral sequence relating the Tomita–Takesaki modular theory of $\mathcal{M}$ to the derived category $D^b_c(\mathrm{Sh}_{\mathbf{k}}(M))$. This is a deep problem connecting operator algebras and microlocal geometry.

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